

Unit 1: Time & Work, Pipes & Cisterns

- Work = Rate $\times$ Time
- Efficiency $\propto 1/\text{Time}$
- If A does work in x days $\rightarrow$ Efficiency = $1/x$
- Combined work: $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$
- Pipes filling/emptying: Net rate = (Inlet rate – Outlet rate)
- Chain Rule: Work ratio = $\frac{\text{Men} \times \text{Days} \times \text{Hours}}{\text{Work}}$
- Together work: $\frac{xy}{x+y}$ days
- Efficiency ratio: If A is k times B $\rightarrow k:1$
- Pipes net time: $\frac{xy}{y-x}$ (when $y > x$)

Statistics

- Mean: $\bar{x} = \frac{\sum x_i}{n}$
- Median: Middle value (odd n) or average of two middle values (even n)
- Mode: Most frequent value
- Standard Deviation: $\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$
- Variance: σ^2
- Normal Distribution: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
- Outlier: Value outside $Q1 - 1.5 \cdot IQR$ or $Q3 + 1.5 \cdot IQR$
- Combined Mean: $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$
- Coefficient of Variation: $CV = \frac{\sigma}{\bar{x}} \times 100\%$
- Z-score: $z = \frac{x - \mu}{\sigma}$

Unit 2: Time, Speed & Distance

1. Distance = Speed $\times$ Time
2. Speed = Distance / Time
3. Average Speed: $\frac{\text{Total Distance}}{\text{Total Time}}$
4. Harmonic Mean (equal distances): $\frac{2xy}{x+y}$
5. Relative Speed: (Same direction $\rightarrow$ difference, opposite $\rightarrow$ sum)
6. Boats: Effective speed = (Boat $\pm$ Stream)
7. Trains: Time = $\frac{\text{Length of train}}{\text{Relative speed}}$
8. Races: Ratio method for time/distance differences
9. Escalator problems: Effective speed = (Person $\pm$ Escalator)

Linear Algebra

- Linearity: $f(ax + by) = af(x) + bf(y)$
- Two-variable system: Solve using substitution/elimination
- Three-variable system: Solve using determinants (Cramer's Rule)
- Cramer's Rule: $x = \frac{\Delta_x}{\Delta}, y = \frac{\Delta_y}{\Delta}, z = \frac{\Delta_z}{\Delta}$
- Determinant (2×2): $|A| = ad - bc$
- Determinant (3×3): Expansion by minors
- Inverse of matrix: $A^{-1} = \frac{1}{|A|} \text{Adj}(A)$

Quadratic Equations & Polynomials

- General form: $ax^2 + bx + c = 0$
- Roots: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Sum of roots = $-b/a$, Product of roots = c/a
- Discriminant: $D = b^2 - 4ac$
 - $D > 0$: real distinct roots
 - $D = 0$: real equal roots
 - $D < 0$: complex roots
- Factor theorem: If $f(a) = 0$, then $(x-a)$ is a factor

Unit 3: Permutation & Combination

- Permutation: $P(n, r) = \frac{n!}{(n-r)!}$
- Combination: $C(n, r) = \frac{n!}{r!(n-r)!}$
- Relation: $P(n, r) = C(n, r) \cdot r!$
- Fundamental Counting Principle: Multiply choices
- Circular Permutations: $(n - 1)!$
- With repetition: n^r
- Combination with repetition: $C(n + r - 1, r)$

Probability

- Probability = $\frac{\text{Favorable outcomes}}{\text{Total outcomes}}$
- Independent Events: $P(A \cap B) = P(A) \cdot P(B)$
- Mutually Exclusive: $P(A \cup B) = P(A) + P(B)$
- Conditional Probability: $P(A | B) = \frac{P(A \cap B)}{P(B)}$
- Bayes' Theorem:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

- Total Probability: $P(B) = \sum P(B | A_i) \cdot P(A_i)$
- Complement rule: $P(A') = 1 - P(A)$

Clocks & Calendars

- Angle between hands: $\theta = |30H - 5.5M|$
- Hands coincide every $65 \frac{5}{11}$ minutes
- Hands at right angle: $\theta = 90^\circ$
- Odd days: Days mod 7
- Leap Year: Divisible by 4 (except centuries not divisible by 400)
- Ordinary year = 365 days = 52 weeks + 1 odd day
- Leap year = 366 days = 52 weeks + 2 odd days

Unit 4: Logarithms, Surds & Indices

- Laws of Logarithms:
 - $\log_a(MN) = \log_a M + \log_a N$
 - $\log_a \frac{M}{N} = \log_a M - \log_a N$
 - $\log_a(M^k) = k \log_a M$
- Change of Base: $\log_a M = \frac{\log_b M}{\log_b a}$
- Surds: $\sqrt[n]{a^m} = a^{m/n}$
- Indices:
 - $a^m \cdot a^n = a^{m+n}$
 - $\frac{a^m}{a^n} = a^{m-n}$
 - $(a^m)^n = a^{mn}$
 - $a^{-n} = \frac{1}{a^n}$
- Special values: $\log_a 1 = 0, \log_a a = 1$
- Surds simplification: $\sqrt{ab} = \sqrt{a}\sqrt{b}$

Logical Reasoning (Conceptual Tools)

- Dice: Opposite faces sum = 7 (standard dice)
- Logic Gates: AND, OR, NOT, NAND, NOR, XOR truth tables
- Blood Relations: Use generation levels (+1 child, -1 parent)
- Puzzles, Caselets, Seating/Floor arrangements → solved by systematic elimination